

PETER CHEGE

Backend Engineer • Data Engineer • AI Systems

+254 713 244 359 • peterchege12@gmail.com • [linkedin.com/in/chege-peter](https://www.linkedin.com/in/chege-peter) • github.com/peter-njoro

SUMMARY

Computer Science graduate (BSc, Second Class Honours — Upper Division) with hands-on experience in backend engineering, cloud infrastructure, and applied AI. I build things end-to-end — from architecture to deployment — and I document them so future-me doesn't inherit a mess. Currently deepening skills in data engineering and agentic systems.

SKILLS

Languages: Python, JavaScript, C++, SQL, Bash
Backend: Django, REST APIs, WebSockets, systemd, nginx
Data: PostgreSQL, SQLite, pandas, data cleaning, statistical analysis
DevOps: Docker, AWS EC2/ECS, ALB, ACM, GitHub Actions (familiar)
AI / ML: OpenCV, face_recognition, sentence-transformers, LLM integration (Groq, Gemini)
Tools: Git, uv, Fish shell, Linux (Arch, Fedora)

PROJECTS

Contexa — Personal Project

November, 2025 – Present

- Built a local-first context engine designed as a persistence and sync layer for AI agents — runs as a background daemon (contexad), stores versioned JSON documents in SQLite, and exposes a local HTTP API.
- Implemented end-to-end encrypted device sync over LAN or a self-hostable WebSocket relay; the relay is stateless and never sees plaintext.
- Shipped a typed Python SDK (pip install contexa), a CLI, and optional semantic search via sentence-transformers.
- Currently writing unit and property-based tests with pytest and Hypothesis.

Stack: Python 3.11, SQLite, uv, Ed25519 encryption, WebSockets

ReconRoll (FaceTrack Lite) — Personal Project

May, 2025 – Present

- Built a facial recognition attendance system for academic environments — lecturers initiate sessions, the system recognizes enrolled students via face encodings, logs attendance, and saves unidentified faces for review.
- Dockerized for reproducible deployments; configured nginx as a reverse proxy with TLS termination.
- Planned migration to AWS ECS with Application Load Balancer and ACM-managed certificates for production-grade HTTPS.

Stack: Python, Django, OpenCV, PostgreSQL, Docker, nginx, AWS

Task & Project Management Tool — Academic Project (GPL-licensed)

January, 2025 — April, 2025

- Built role-based access control, a real-time notification system (project events, role changes, mentions), and AJAX-powered search in Django.
- Reduced task processing time by 40% through query optimisation and workflow improvements.

Stack: Python, Django, PostgreSQL, JavaScript

EXPERIENCE

Data Science Fellow — ALX Africa (Remote) July 2025 – Present

- Completed structured training in data analytics and Python for data science, covering data cleaning, exploratory analysis, and statistical reasoning.
- Worked on real-world datasets in a peer-collaborative sprint environment with version-controlled deliverables.

EDUCATION

BSc Computer Science — Second Class Honours, Upper Division — Gretsä University Sep 2022 – Oct 2025

Relevant coursework: Database Systems, API Development, Software Engineering Principles, Algorithms.

Data Analytics & Python Courses — ALX Africa 2025

INTERESTS

AI & Agentic Systems • Open Source • Machine Vision • Linux • Anime